BoB Navier Stokes Pressure Gradient till first week of December 2026 - ChatGPT AI model computation and graphics

From: Srinivasan Kannan <ka.shrinivaasan@gmail.com>
To: Krishna iResearch - Srinivasan Kannan (also known as: Ka.Shrinivaasan, Shrinivas Kannan) <ksrinivasan@krishna-iresearch.org>
Date: 2026-10-16 11:39

---------- Forwarded message ---------
From: **Srinivasan Kannan** <ka.shrinivaasan@gmail.com>
Date: Fri, Oct 16, 2026 at 11:24 AM
Subject: Re: GIF from ka shrinivaasan
To: ka.shreenivaasan <ka.shreenivaasan@gmail.com>, Krishna iResearch - Srinivasan Kannan (also known as: Ka.Shrinivaasan, Shrinivas Kannan) <ksrinivasan@krishna-iresearch.org>

Worked for 18s

Generated both files for 6 October–6 December 2026:

- Download the GIS-style PNG — 15 Oct 2026
- Download the 2-month animated GIF

The animation shows pressure-gradient magnitude as the field and pressure-gradient acceleration direction as vectors, with isobars overlaid.

Important: the 12–15 October disturbance is anchored to the current IMD guidance; beyond the useful forecast range, the animation transitions to an ensemble/climatological scenario rather than claiming deterministic weather prediction. IMD currently identifies a possible low-pressure development around 12–15 October. ECMWF's deterministic pressure/wind product itself is a medium-range product extending to about 15 days.

If you want, I can next , produce a **much more rigorous 0.25° × 0.25° Navier–Stokes pressure-gradient model**, for the Bay of Bengal. ❶

NeuronRain Documentation and Licensing: http://neuronrain-documentation.readthedocs.io/en/latest/
Personal website(research): https://acadpdrafts.readthedocs.io/en/latest/

On Fri, 16 Oct, 2026, 11:24 am Srinivasan Kannan, <ka.shrinivaasan@gmail.com> wrote:

> Yes. I can construct a **Bay of Bengal pressure-gradient map for the next 2 months (6 October–6 December 2026)**, but there is an important distinction:

A true Navier–Stokes forecast requires atmospheric/ocean initial conditions and numerical integration. A deterministic day-by-day pressure-gradient map 60 days ahead is **not physically predictable at weather-model resolution**. ECMWF notes that detailed atmospheric predictability is limited beyond roughly the medium-range; seasonal forecasts instead describe statistical departures from normal.

What I can compute

I would calculate the horizontal pressure-gradient field

$$
\boxed{\nabla_h p =
\left(\frac{\partial p}{\partial x},
\frac{\partial p}{\partial y}\right)}
$$

and, for an atmospheric model, the pressure-gradient acceleration

$$
\boxed{\mathbf a_p=-\frac{1}{\rho}\nabla_h p}
$$

over approximately **5–22°N, 78–100°E**, covering the entire Bay of Bengal.

For the next two months I'd produce:

Period	Appropriate product
6–14 Oct 2026	Numerical/extended-range pressure-gradient field
15–31 Oct	Extended/ensemble tendency
Nov 2026	Seasonal/ensemble anomaly field
1–6 Dec	Seasonal/ensemble anomaly field

The current IMD sea-area bulletin provides actual observations/analysis for the Bay. On 6 October it reports generally weak winds of 5–10 kt over much of the Bay, with 10–15 kt in the northwest Bay. IMD is also currently issuing an extended-range forecast through 14 October.

The **October–November period is particularly important** because Bay of Bengal post-monsoon low-pressure systems and cyclones become climatologically favored.

Best version of the map

I recommend constructing a **Navier–Stokes-derived ensemble pressure-gradient map**, rather than pretending there is a deterministic 60-day forecast:

Input

- ECMWF/ERA5 or equivalent MSLP
- 850-hPa winds
- SST
- surface pressure
- geopotential height
- IMD observations
- seasonal ENSO/IOD state

Output

- pressure \(p(x,y,t)\)
- \(∂p/∂x\)
- \(∂p/∂y\)
- \(|∇p|\)
- pressure-gradient acceleration \(-∇p/ρ\)
- wind vectors
- low-pressure/high-pressure centers
- cyclone-favorable regions
- uncertainty/ensemble spread

ECMWF specifically provides MSLP and 850-hPa wind fields, which are suitable for deriving the pressure-gradient component.

, with and . For the 60-day portion, it would be an **ensemble/climatological projection**, not a claimed deterministic cyclone forecast.

NeuronRain Documentation and Licensing: http://neuronrain-documentation.readthedocs.io/en/latest/
Personal website(research): https://acadpdrafts.readthedocs.io/en/latest/

On Fri, 16 Oct, 2026, 9:47 am Srinivasan Kannan, <ka.shrinivaasan@gmail.com> wrote:

> ChatGPT Navier Stokes pressure gradient animation upto December 2026 first week
>
> ---
> Personal website(research): https://acadpdrafts.readthedocs.io/en/latest/
> NeuronRain : http://neuronrain-documentation.readthedocs.io/en/latest/
> ---